공학수학(2)

# 14.6 Triple Integrals

# Triple Integrals over Rectangular Boxes

Let's first deal with the simplest case where $f$ is defined on a rectangular box:

$$\mathbf{1} \qquad B = \{(x, y, z) \mid a \le x \le b,\ c \le y \le d,\ r \le z \le s\}$$

The first step is to divide $B$ into sub-boxes.

We do this by dividing the interval $[a, b]$ into $l$ subintervals $[x_{i-1}, x_i]$ of equal width $\Delta x$, dividing $[c, d]$ into $m$ subintervals of width $\Delta y$, and dividing $[r, s]$ into $n$ subintervals of width $\Delta z$.

# Triple Integrals over Rectangular Boxes

The planes through the endpoints of these subintervals parallel to the coordinate planes divide the box $B$ into $lmn$ sub-boxes

$$B_{ijk} = [x_{i-1},\ x_i] \times [y_{j-1},\ y_j] \times [z_{k-1},\ z_k]$$

which are shown in Figure 1. Each sub-box has volume $\Delta V = \Delta x\, \Delta y\, \Delta z$.

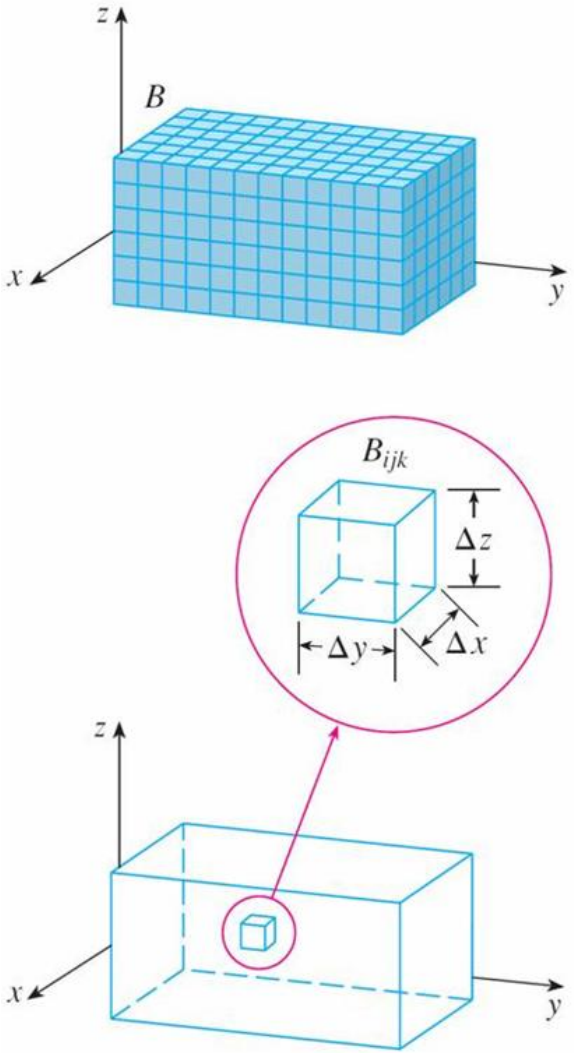



**Figure 1**

# Triple Integrals over Rectangular Boxes

Then we form the **triple Riemann sum**

$$\textbf{2} \qquad \sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(x_{ijk}^*,\ y_{ijk}^*,\ z_{ijk}^*)\ \Delta V$$

where the sample point $(x_{ijk}^*,\ y_{ijk}^*,\ z_{ijk}^*)$ is in $B_{ijk}$.

By analogy with the definition of a double integral, we define the triple integral as the limit of the triple Riemann sums in (2).

# Triple Integrals over Rectangular Boxes

**3 Definition** The **triple integral** of $f$ over the box $B$ is

$$\iiint_B f(x,y,z)\,dV = \lim_{l,m,n\to\infty} \sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(x_{ijk}^*,\ y_{ijk}^*,\ z_{ijk}^*)\,\Delta V$$

if this limit exists.

Again, the triple integral always exists if $f$ is continuous.

We can choose the sample point to be any point in the sub-box, but if we choose it to be the point $(x_i, y_j, z_k)$ we get a simpler-looking expression for the triple integral:

$$\iiint_B f(x,y,z)\,dV = \lim_{l,m,n\to\infty} \sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(x_i,\ y_j,\ z_k)\,\Delta V$$

# Triple Integrals over Rectangular Boxes

Just as for double integrals, the practical method for evaluating triple integrals is to express them as iterated integrals as follows.

**4 Fubini's Theorem for Triple Integrals** If $f$ is continuous on the rectangular box $B = [a, b] \times [c, d] \times [r, s]$, then

$$\iiint_B f(x, y, z)\, dV = \int_r^s \int_c^d \int_a^b f(x, y, z)\, dx\, dy\, dz$$

The iterated integral on the right side of Fubini's Theorem means that we integrate first with respect to $x$ (keeping $y$ and $z$ fixed), then we integrate with respect to $y$ (keeping $z$ fixed), and finally we integrate with respect to $z$.

# Triple Integrals over Rectangular Boxes

There are five other possible orders in which we can integrate, all of which give the same value.

For instance, if we integrate with respect to $y$, then $z$, and then $x$, we have

$$\iiint_B f(x,\ y,\ z)\ dV = \int_a^b \int_r^s \int_c^d f(x,\ y,\ z)\ dy\ dz\ dx$$

# Example 1

Evaluate the triple integral $\iiint_B xyz^2 \, dV$, where $B$ is the rectangular box given by

$$B = \{(x, y, z) \mid 0 \le x \le 1, \ -1 \le y \le 2, \ 0 \le z \le 3\}$$

Solution:

We could use any of the six possible orders of integration.

If we choose to integrate with respect to $x$, then $y$, and then $z$, we obtain

$$\iiint_B xyz^2 \, dV = \int_0^3 \int_{-1}^2 \int_0^1 xyz^2 \, dx \, dy \, dz$$

# Triple Integrals over General Regions

Now we define the **triple integral over a general bounded region *E*** in three-dimensional space (a solid) by much the same procedure that we used for double integrals.

We enclose *E* in a box *B* of the type given by Equation 1. Then we define *F* so that it agrees with *f* on *E* but is 0 for points in *B* that are outside *E*.

By definition,

$$\iiint_E f(x, y, z)\, dV = \iiint_B F(x, y, z)\, dV$$

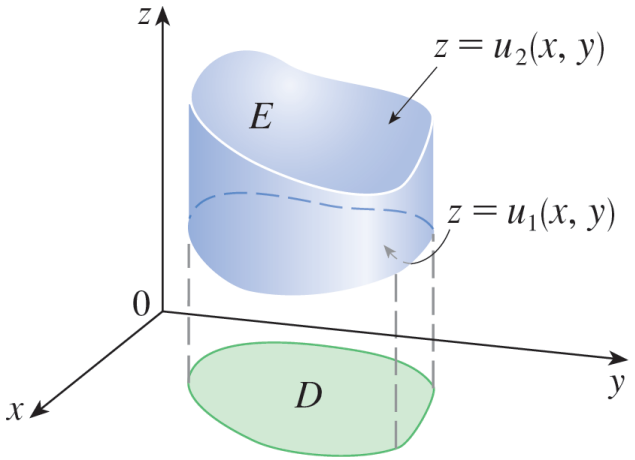
This integral exists if *f* is continuous and the boundary of *E* is "reasonably smooth."

# Triple Integrals over General Regions

A solid region $E$ is said to be of **type 1** if it lies between the graphs of two continuous functions of $x$ and $y$, that is,

$$\mathbf{5} \quad E = \{(x, y, z) \mid (x, y) \in D,\ u_1(x, y) \le z \le u_2(x, y)\}$$

where $D$ is the projection of $E$ onto the $xy$-plane as shown in Figure 2.




A type 1 solid region

**Figure 2**

# Triple Integrals over General Regions

Notice that the upper boundary of the solid $E$ is the surface with equation $z = u_2(x, y)$, while the lower boundary is the surface $z = u_1(x, y)$.

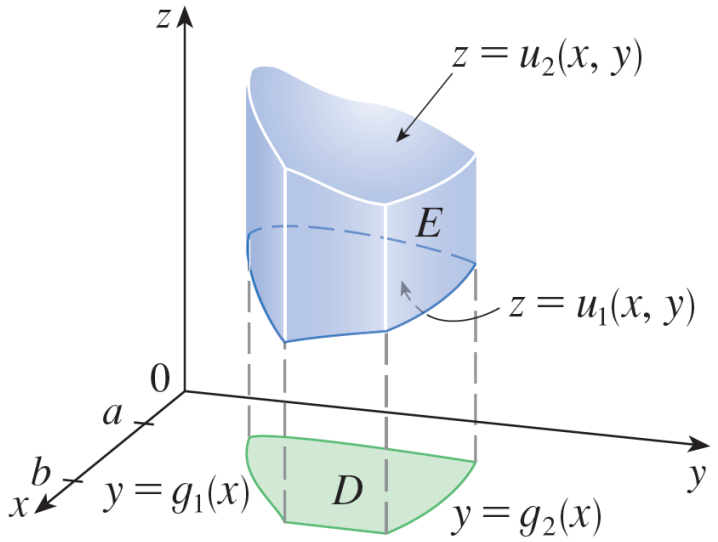
By the same sort of argument, it can be shown that if $E$ is a type 1 region given by Equation 5, then

$$\textbf{6} \qquad \iiint_E f(x, y, z)\, dV = \iint_D \left[ \int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z)\, dz \right] dA$$

The meaning of the inner integral on the right side of Equation 6 is that $x$ and $y$ are held fixed, and therefore $u_1(x, y)$ and $u_2(x, y)$ are regarded as constants, while $f(x, y, z)$ is integrated with respect to $z$.

# Triple Integrals over General Regions

In particular, if the projection $D$ of $E$ onto the $xy$-plane is a type I plane region (as in Figure 3).




A type 1 solid region where the projection $D$ is a type I plane region

**Figure 3**

# Triple Integrals over General Regions

Then

$$E = \{(x,\ y,\ z) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x),\ u_1(x,\ y) \le z \le u_2(x,\ y)\}$$

and Equation 6 becomes

$$\textbf{7} \qquad \iiint_E f(x,\ y,\ z)\ dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y)}^{u_2(x,y)} f(x,\ y,\ z)\ dz\ dy\ dx$$

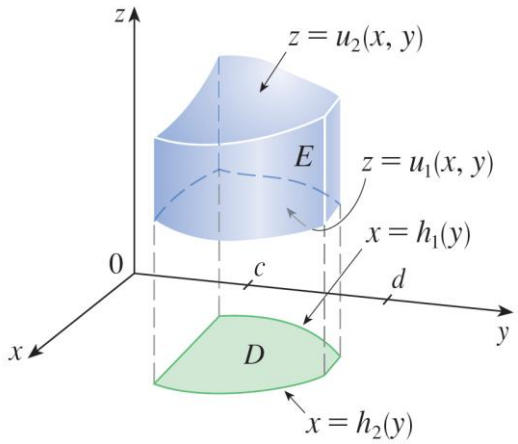
# Triple Integrals over General Regions

If, on the other hand, $D$ is a type II plane region (as in Figure 4), then

$$E = \{(x, y, z) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y),\ u_1(x, y) \le z \le u_2(x, y)\}$$

and Equation 6 becomes

$$\text{8} \qquad \iiint_E f(x, y, z)\, dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z)\, dz\, dx\, dy$$




A type 1 solid region with a type II projection

**Figure 4**

# Example 2

Evaluate $\iiint_E z\, dV$ where $E$ is the solid in the first octant bounded by the surface $z = 12xy$ and the planes $y = x$ , $x = 1$.

## Solution:

When we set up a triple integral it's wise to draw *two* diagrams: one of the solid region $E$ (Figure 5) and, for a type 1 region, one of its projection $D$ onto the $xy$-plane (Figure 6).

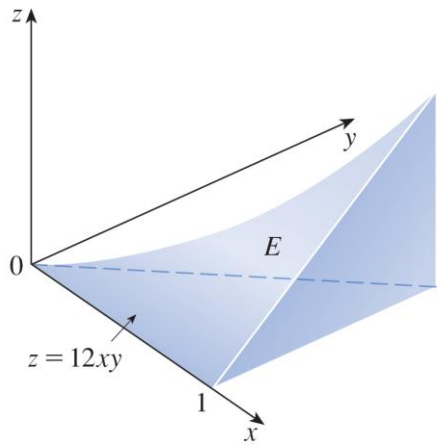



Figure 5

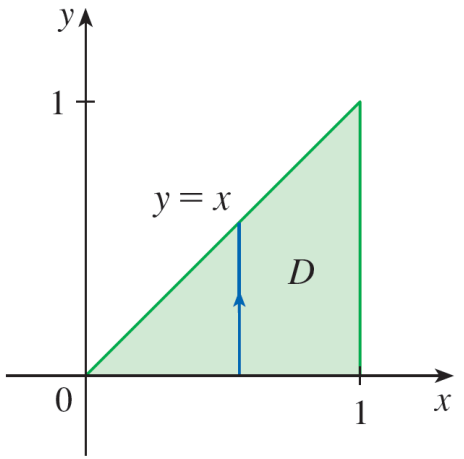



Figure 6

# Example 2 – Solution

The lower boundary of the solid $E$ is the plane $z = 0$ and the upper boundary is the surface $z = 12xy$, so we use $u_1(x, y) = 0$ and $u_2(x, y) = 12xy$ in Formula 7.

Notice that the projection of $E$ onto the $xy$-plane is the triangular region shown in Figure 6, and we have

$$E = \{(x, y, z) \mid 0 \le x \le 1, 0 \le y \le x, 0 \le z \le 12xy\} \quad \textbf{9}$$

# Triple Integrals over General Regions

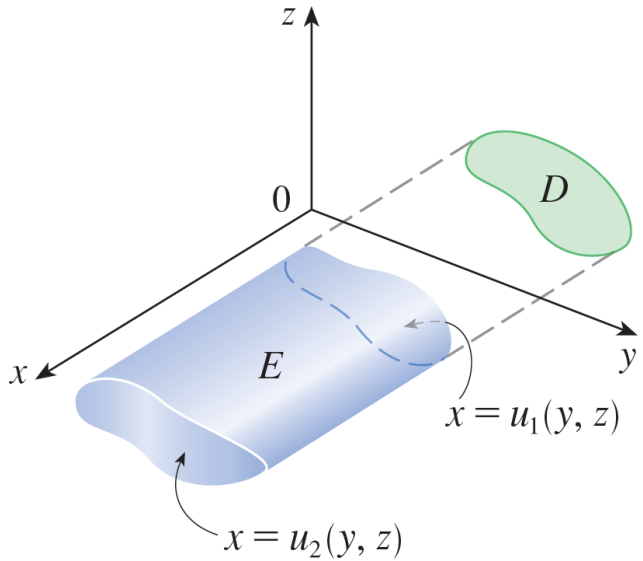
A solid region $E$ is of **type 2** if it is of the form

$$E = \{(x, y, z) \mid (y, z) \in D,\ u_1(y, z) \le x \le u_2(y, z)\}$$

where, this time, $D$ is the projection of $E$ onto the $yz$-plane (see Figure 8).

The back surface is $x = u_1(y, z)$, the front surface is $x = u_2(y, z)$, and we have

**10**
$$\iiint_E f(x, y, z)\, dV = \iint_D \left[ \int_{u_1(y,z)}^{u_2(y,z)} f(x, y, z)\, dx \right] dA$$




A type 2 region

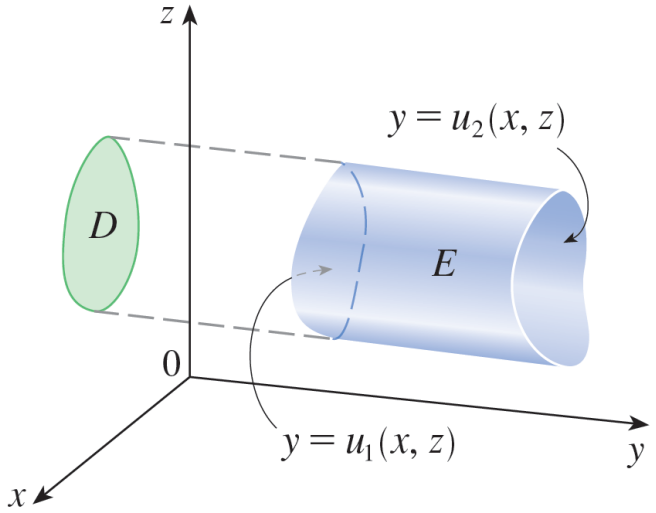
**Figure 8**

# Triple Integrals over General Regions

Finally, a **type 3** region is of the form

$$E = \{(x, y, z) \mid (x, z) \in D,\ u_1(x, z) \le y \le u_2(x, z)\}$$

where $D$ is the projection of $E$ onto the $xz$-plane, $y = u_1(x, z)$ is the left surface, and $y = u_2(x, z)$ is the right surface (see Figure 9).




A type 3 region

**Figure 9**

# Triple Integrals over General Regions

For this type of region we have

$$\textbf{11} \qquad \iiint_E f(x,\ y,\ z)\ dV = \iint_D \left[ \int_{u_1(x,z)}^{u_2(x,z)} f(x,\ y,\ z)dy \right] dA$$

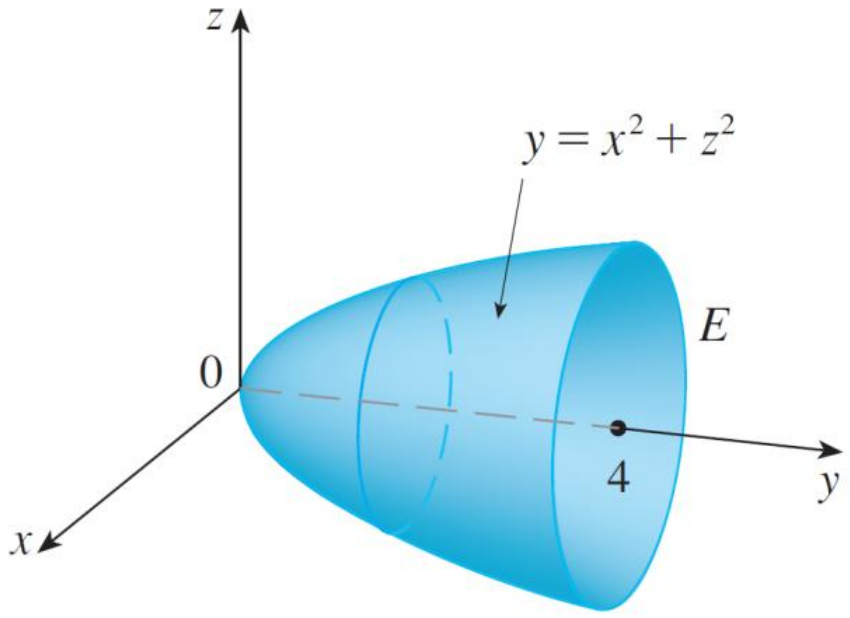
In each of Equations 10 and 11 there may be two possible expressions for the integral depending on whether $D$ is a type I or type II plane region (and corresponding to Equations 7 and 8).

# Example 3

Evaluate $\iiint_E \sqrt{x^2 + z^2}\, dV$, where $E$ is the region bounded by the paraboloid $y = x^2 + z^2$ and the plane $y = 4$.

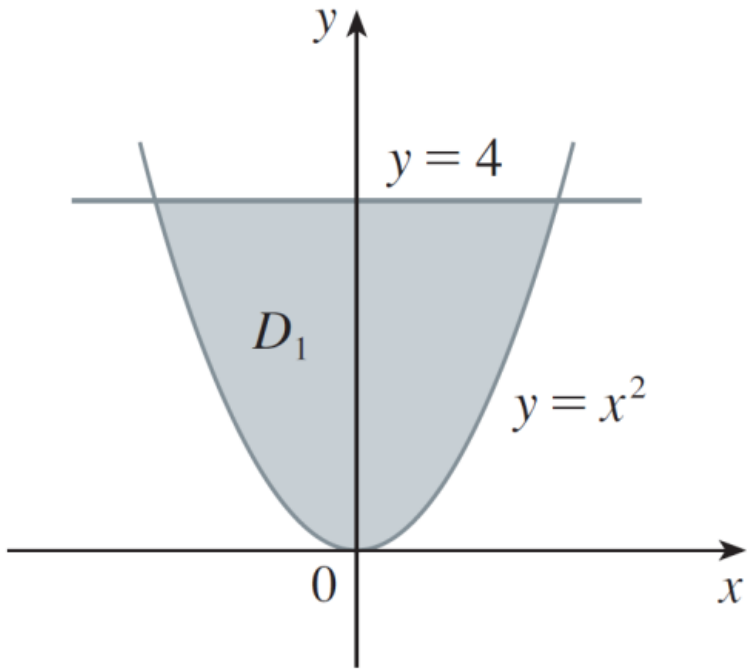
Solution:

The solid $E$ is shown in Figure 10.



# Example 3

If we regard it as a type 1 region, then we need to consider its projection $D_1$ onto the *xy*-plane, which is the parabolic region in Figure 11. (The trace of $y = x^2 + z^2$ in the plane $z = 0$ is the parabola $y = x^2$.)



# Example 3

From $y = x^2 + z^2$ we obtain $z = \pm\sqrt{y - x^2}$, so the lower boundary surface of $E$ is $z = -\sqrt{y - x^2}$ and the upper surface is $z = \sqrt{y - x^2}$.

Therefore the description of $E$ as a type 1 region is

$$E = \{(x, y, z) \mid -2 \leq x \leq 2,\ x^2 \leq y \leq 4,\ -\sqrt{y - x^2} \leq z \leq \sqrt{y - x^2}\}$$

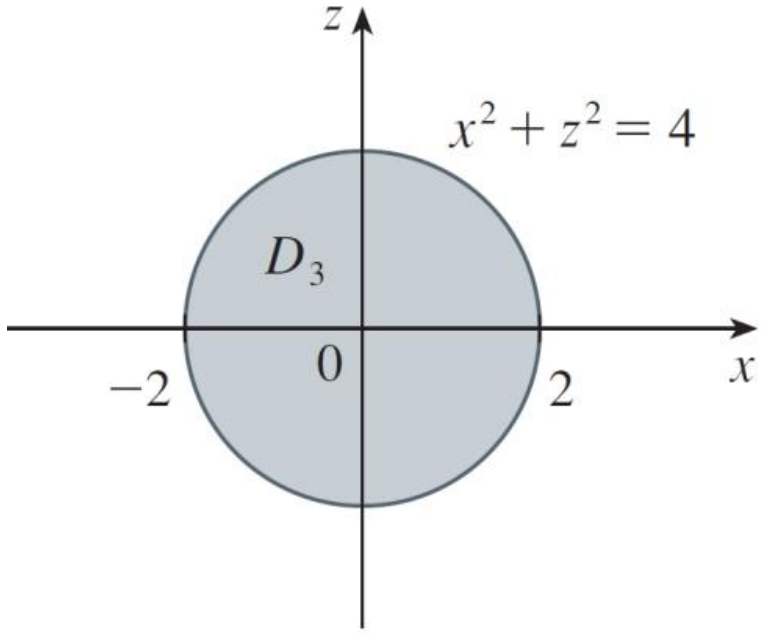
and so we obtain

$$\iiint_E \sqrt{x^2 + z^2}\, dV = \int_{-2}^{2} \int_{x^2}^{4} \int_{-\sqrt{y-x^2}}^{\sqrt{y-x^2}} \sqrt{x^2 + z^2}\, dz\, dy\, dx$$

# Example 3

Although this expression is correct, it is extremely difficult to evaluate. So let's instead consider *E* as a type 3 region.

As such, its projection $D_3$ onto the *xz*-plane is the disk $x^2 + z^2 \leq 4$ shown in Figure 13.



# Example 3

Then the left boundary of *E* is the paraboloid $y = x^2 + z^2$ and the right boundary is the plane $y = 4$, so taking $u_1(x, z) = x^2 + z^2$ and $u_2(x, z) = 4$ in Equation 11, we have

$$\iiint_E \sqrt{x^2 + z^2}\, dV = \iint_{D_3} \left[ \int_{x^2+z^2}^{4} \sqrt{x^2 + z^2}\, dy \right] dA$$

$$= \iint_{D_3} (4 - x^2 - z^2)\sqrt{x^2 + z^2}\, dA$$

# Changing the Order of Integration

Fubini's Theorem for Triple Integrals allows us to express a triple integral as an iterated integral, and there are six different orders of integration in which we can do this.

Given an iterated integral, it may be advantageous to change the order of integration because evaluating an iterated integral in one order may be simpler than in another.

In the next example we investigate equivalent iterated integrals using different orders of integration.

# Example 4

Express the iterated integral $\int_0^1 \int_0^{x^2} \int_0^y f(x,y,z)\, dz\, dy\, dx$ as a triple integral and then rewrite it as an iterated integral in the following orders.

(a) Integrate first with respect to $x$, then $z$, and then $y$.

(b) Integrate first with respect to $y$, then $x$, and then $z$.

Solution:

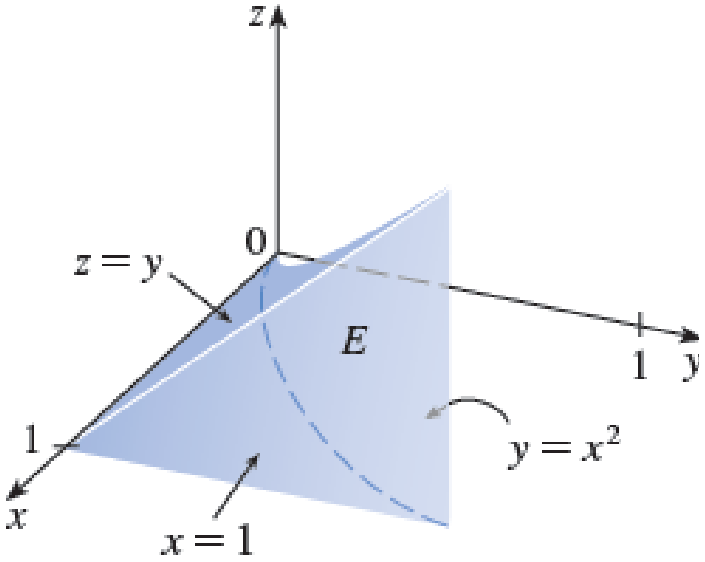
We can write

$$\int_0^1 \int_0^{x^2} \int_0^y f(x,y,z)\, dz\, dy\, dx = \iiint_E f(x,y,z)\, dv$$

where $E = \{(x,y,z) \mid 0 \le x \le 1, 0 \le y \le x^2, 0 \le z \le y\}$.

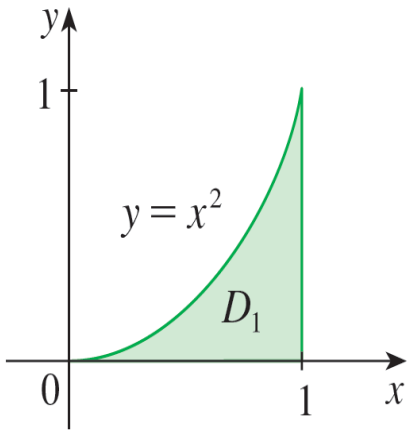
# Example 4 – Solution

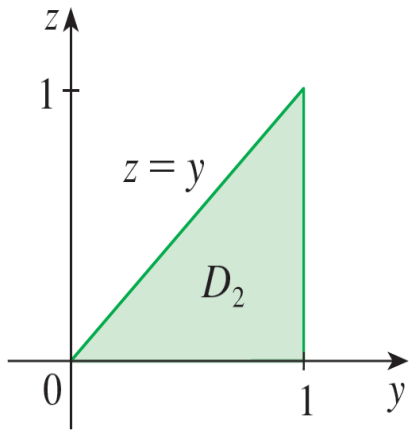
From this description of $E$ as a type 1 region we see that $E$ lies between the lower surface $z = 0$ and the upper surface $z = y$, and its projection onto the $xy$-plane is $\{(x, y) \mid 0 \le x \le 1, 0 \le y \le x^2\}$, as shown in Figures 14 and 15.

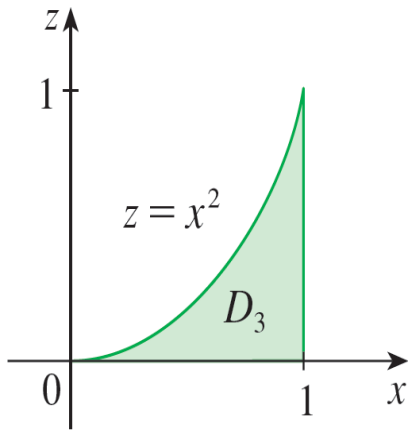



The solid $E$

**Figure 14**




Projections of $E$

**Figure 15**

# Example 4 – Solution

So $E$ is the solid enclosed by the planes $z = 0$, $x = 1$, $y = z$ and the parabolic cylinder $y = x^2$ (or $x = \sqrt{y}$).

Using Figure 14 as a guide, we can write projections onto the three coordinate planes as follows (see Figure 15):

onto the $xy$-plane:
$$D_1 = \{(x, y) \mid 0 \le x \le 1, 0 \le y \le x^2\}$$
$$= \{(x, y) \mid 0 \le y \le 1, \sqrt{y} \le x \le 1\}$$

onto the $yz$-plane:
$$D_2 = \{(y, z) \mid 0 \le y \le 1, 0 \le z \le y\}$$
$$= \{(y, z) \mid 0 \le z \le 1, z \le y \le 1\}$$

onto the $xz$-plane:
$$D_3 = \{(x, z) \mid 0 \le x \le 1, 0 \le z \le x^2\}$$
$$= \{(x, z) \mid 0 \le z \le 1, \sqrt{z} \le x \le 1\}$$

# Example 4 – Solution

(a) In order to integrate first with respect to $x$, then $z$, and then $y$, we need to consider $E$ as a type 2 region where the back boundary is the surface $x = \sqrt{y}$ and the front boundary is the plane $x = 1$; the projection onto the $yz$-plane is $D_2$.

We describe $E$ by

$$E = \{(x, y, z) \mid 0 \le y \le 1, 0 \le z \le y, \sqrt{y} \le x \le 1\}$$

and then

$$\iiint_E f(x, y, z)\, dv = \int_0^1 \int_0^y \int_{\sqrt{y}}^1 f(x, y, z)\, dx\, dz\, dy$$

# Example 4 – Solution

b) In order to integrate first with respect to $y$, then $x$, and then $z$, we need to consider $E$ as a type 3 region where the left boundary is the plane $y = z$ and the right boundary is the surface $y = x^2$.

The projection onto the $xz$-plane is $D_3$ and

$$E = \{(x, y, z) \mid 0 \le z \le 1, \sqrt{z} \le x \le 1, z \le y \le x^2\}$$

Thus

$$\iiint_E f(x, y, z)\, dv = \int_0^1 \int_{\sqrt{z}}^1 \int_z^{x^2} f(x, y, z)\, dy\, dx\, dz$$

# Applications of Triple Integrals

We know that if $f(x) \geq 0$, then the single integral $\int_a^b f(x)\,dx$ represents the area under the curve $y = f(x)$ from $a$ to $b$, and if $f(x, y) \geq 0$, then the double integral $\iint_D f(x, y)\,dA$ represents the volume under the surface $z = f(x, y)$ and above $D$.

The corresponding interpretation of a triple integral $\iiint_E f(x, y, z)\,dV$, where $f(x, y, z) \geq 0$, is not very useful because it would be the "hypervolume" of a four-dimensional object and, of course, that is very difficult to visualize. (Remember that $E$ is just the *domain* of the function $f$; the graph of $f$ lies in four-dimensional space.)

# Applications of Triple Integrals

Nonetheless, the triple integral $\iiint_E f(x, y, z)\, dV$ can be interpreted in different ways in different physical situations, depending on the physical interpretations of $x$, $y$, $z$ and $f(x, y, z)$.

Let's begin with the special case where $f(x, y, z) = 1$ for all points in $E$. Then the triple integral does represent the volume of $E$:

**12** $$V(E) = \iiint_E dV$$

# Example 5

Use a triple integral to find the volume of the tetrahedron $T$ bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$.

Solution:

The tetrahedron $T$ and its projection $D$ onto the $xy$-plane are shown in Figures 16 and 17.

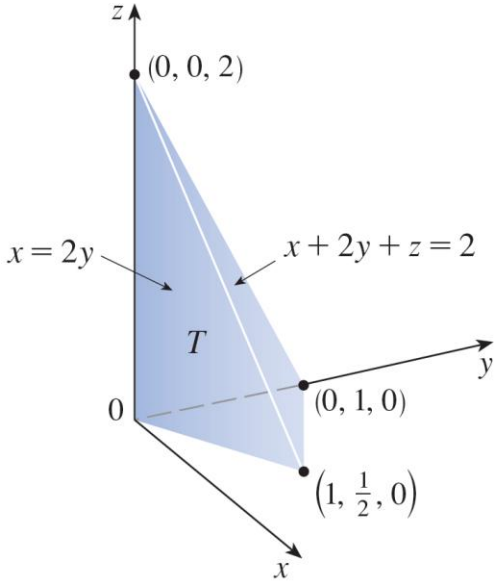



Figure 16

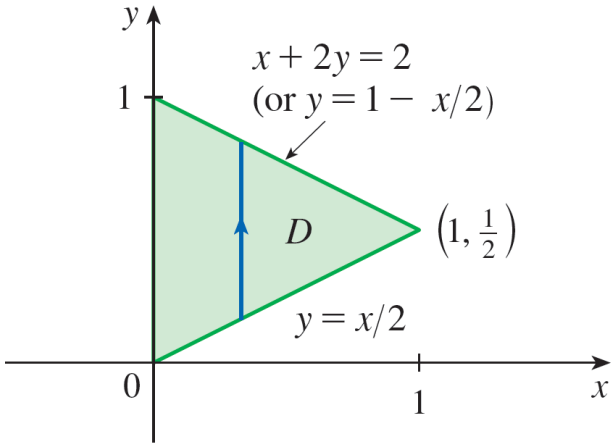



Figure 17

# Example 5 – Solution

The lower boundary of $T$ is the plane $z = 0$ and the upper boundary is the plane $x + 2y + z = 2$, that is, $z = 2 - x - 2y$.

Therefore we have

$$V(T) = \iiint_T dV$$

$$= \int_0^1 \int_{\frac{x}{2}}^{1-\frac{x}{2}} \int_0^{2-x-2y} dz\ dy\ dx$$

$$= \int_0^1 \int_{\frac{x}{2}}^{1-\frac{x}{2}} (2 - x - 2y)\ dy\ dx = \frac{1}{3}$$

(Notice that it is not necessary to use triple integrals to compute volumes. They simply give an alternative method for setting up the calculation.)